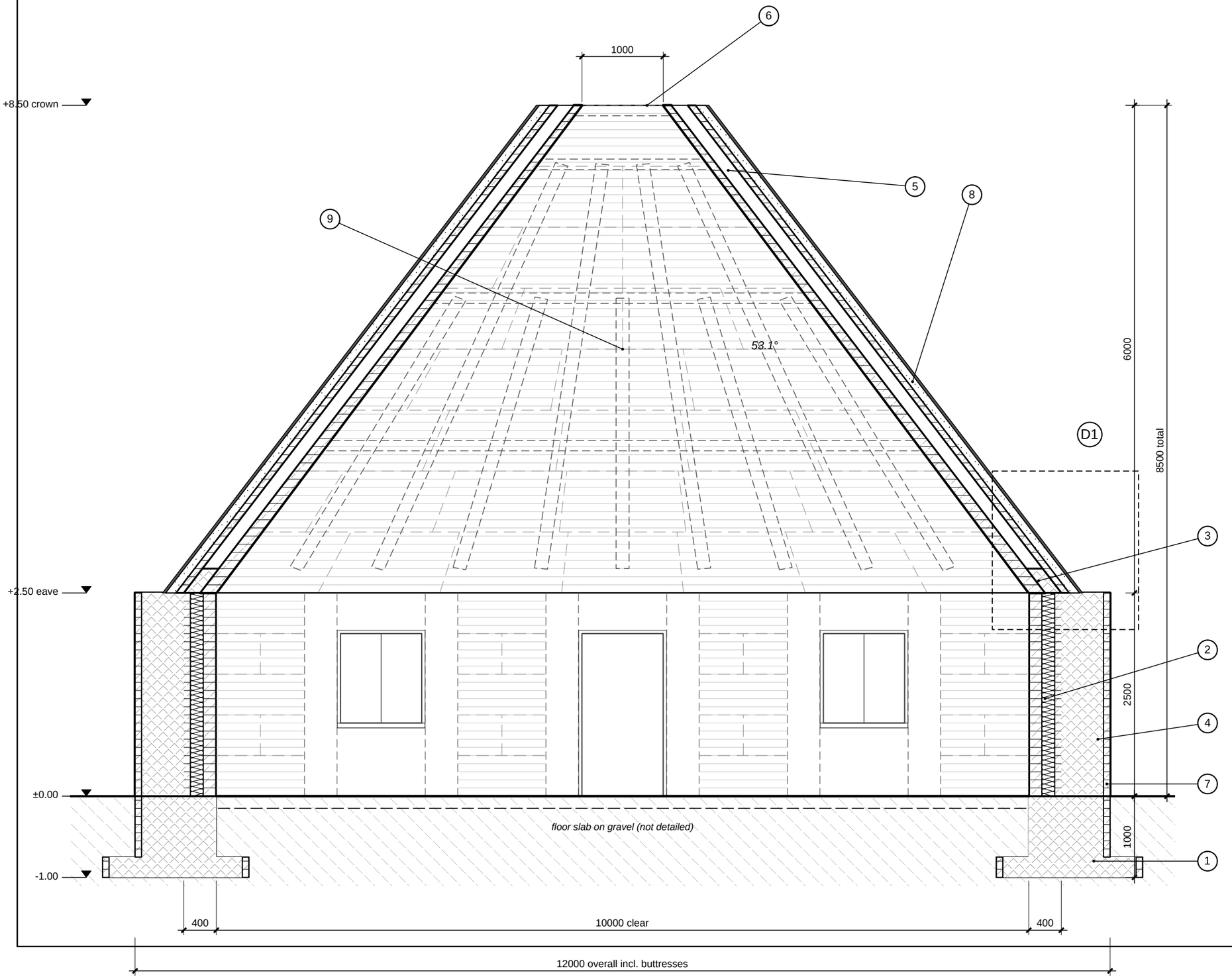
Total section A-A · grid A

through the east and west buttresses, looking north



1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses): printed edge formwork, cast core.
2. Printed wall 400, same pattern as the roof: two 80 inner layers, a 160 air zone, an 80 outer layer.
3. Solid base of the roof: over the first 300 (vertical) above the eave the 160 air zone is filled with cast concrete between the printed layers, so the 400 is solid; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); printed formwork with cast core, rises to +2.50.
5. Roof shell: 400 thick square to the slope at the base (two 80 inner layers, a 160 air zone, an 80 outer layer), 320 at the crown; about 500 / 400 horizontally.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed: mortar beads in 10 mm layers (joints drawn every 100) form permanent formwork. Where a member is solid (roof base, buttress, footing, ribs) the printed shell is filled with cast concrete, crossed hatch. Single hatch = printed only.
8. Outside the printed shell: insulation 100 (dotted), then the outer layer (cladding 30, dark).
9. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid A, ruled-surface columns). The cut lies in the air zone between ribs; rib sections are on the roof-grid sheets.

● from your sketches
○ assumed by me, to confirm

Notes

Layer pattern from your drawing; at the crown the second inner layer ends, so the leaf is 80 there (assumed).

The cut runs through the 4th buttress from the south (y = -743). Beyond it the north wall is seen from inside: every other bay is printed wall, the bays in between are openings (door in the middle, windows either side); buttresses behind the wall are dashed.

Faint lines = printed layer joints (every 100 drawn, real 10); dashed lines on the back face and the wall = divisions between printed pieces in running bond (assumed about 1500 wide).

0 1 2 3 4 5 m

Scale 1:50 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered